

Angle Modulation

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Single Tone Frequency

Modulation

□ Single Tone Modulating/Message Signal, $m(t) = V_m \cos(2\pi f_m t)$

□ Carrier Signal, $c(t) = V_c \sin(2\pi f_c t)$

□ Instantaneous Frequency of FM Signal:

$$f_i(t) = f_c + k_f m(t)$$

$$\Rightarrow f_i(t) = f_c + k_f V_m \cos(2\pi f_m t)$$

$$\Rightarrow \mathbf{f_i(t) = f_c + \Delta f \cos(2\pi f_m t)}$$
 [Frequency Deviation, Δf (for single tone) = $k_f V_m$]

□ Maximum Frequency Deviation or Frequency Deviation = $\mathbf{\max(|f_i(t) - f_c|) = \Delta f}$

□ When $\cos(2\pi f_m t) = +1$, $f_i(t) = f_c + \Delta f = f_{max}$ [Maximum Frequency: When message is positive peak $\rightarrow$ frequency is highest]

□ When $\cos(2\pi f_m t) = -1$, $f_i(t) = f_c - \Delta f = f_{min}$ [Minimum Frequency: When message is negative peak $\rightarrow$ frequency is lowest]

□ When message = 0 $\rightarrow$ frequency = f_c

Single Tone Frequency

□ Instantaneous Phase:

Modulation

$$\theta_i(t) = \int_0^t \omega_i(t) dt = \int_0^t 2\pi f_i(t) dt = 2\pi \int_0^t (f_c + k_f m(t)) dt$$

$$= 2\pi f_c \int_0^t dt + 2\pi k_f \int_0^t m(t) dt$$

$$= 2\pi f_c t + 2\pi k_f \int_0^t V_m \cos(2\pi f_m t) dt$$

$$= \omega_c t + \frac{2\pi k_f V_m \sin(2\pi f_m t)}{2\pi f_m}$$

$$= \omega_c t + \frac{k_f V_m}{f_m} \sin(2\pi f_m t)$$

$$= \omega_c t + \frac{\Delta f}{f_m} \sin(2\pi f_m t) \text{ [as } \Delta f \text{ (for single tone)} = k_f V_m]$$

$$= \omega_c t + m_f \sin(2\pi f_m t) \text{ [Modulation index, } m_f = \frac{\Delta f}{f_m} = \frac{k_f V_m}{f_m}]$$

FM Signal, $\mathbf{Y_{FM}(t) = \phi_{FM}(t) = V_c \sin[\omega_c t + m_f \sin(2\pi f_m t)]}$

Single Tone Frequency Modulation

- ❑ **Deviation Ratio:** Deviation ratio is the ratio of the **maximum frequency deviation** to the **maximum modulating frequency**.

$$\text{Deviation Ratio} = \frac{\Delta f}{f_{m(max)}}$$

- ❑ For single tone FM, $f_{m(max)} = f_m$; So, $\text{Deviation Ratio} = \frac{\Delta f}{f_m} = m_f = \text{Modulation index}$
- ❑ **Maximum Allowed Frequency Deviation (FM), Δf_{max}** : It is the largest frequency deviation allowed by a system or standard, beyond which distortion or interference occurs.

$$\Delta f_{max} = \text{maximum permitted frequency swing from } f_c$$

- ❑ For FM Broadcasting (Wideband FM), $\Delta f_{max} = 75\text{kHz}$
- ❑ For Narrowband FM (Communication systems), $\Delta f_{max} \approx 5\text{kHz}$
- ❑ % Modulation = $\left(\frac{\Delta f}{\Delta f_{max}} \right) \times 100\%$ [How much of allowed deviation is being used]
- ❑ Carson's Rule: 98% of the frequency is covered by $2(\Delta f + f_{m(max)}) \Rightarrow B/W$

Frequency Deviation and Phase Deviation

□ **Frequency Deviation:** $\Delta f = \max(f_i(t) - f_c) = f_i(t)_{max} - f_c \dots \dots \dots (Eqn1)$

□ **Phase Deviation:** $\Delta \theta = \max(\theta_i(t) - \theta_c) = \theta_i(t)_{max} - \theta_c$

□ For single tone FM, why $\Delta f = k_f V_m$? [k_f = frequency sensitivity, V_m = Amplitude of $m(t)$]

We know, Instantaneous Frequency of FM Signal:

$$f_i(t) = f_c + k_f m(t)$$

$$\Rightarrow f_i(t) = f_c + k_f V_m \cos(2\pi f_m t) [m(t) = V_m \cos(2\pi f_m t); \text{single tone modulating signal}]$$

If $\cos(2\pi f_m t) = 1$, we get

$$f_i(t)_{max} = f_c + k_f V_m$$

$$\Rightarrow f_i(t)_{max} - f_c = k_f V_m$$

$$\Rightarrow \Delta f = k_f V_m [\text{From Eqn1}]$$

Examples of Frequency Modulation

Example 1 Determine the percentage modulation for an FM wave with a frequency deviation of 10 kHz, if the maximum deviation allowed is 25 kHz.

$$\rightarrow \% \text{ Modulation} = \frac{\Delta f}{\Delta f_{\max}} \times 100$$

$$= \frac{10}{25} \times 100$$

$$= 40 \%$$

Example 2. In an FM system, if the maximum value of deviation is 75 kHz and the maximum modulating frequency is 10 kHz, calculate the deviation ratio and bandwidth of the system using the Carson rule.

$$\sim \Delta f_{\max} = 75 \text{ kHz}$$

$$f_{m(\max)} = 10 \text{ kHz}$$

$$\rightarrow \text{Deviation ratio} = \frac{\Delta f_{\max}}{f_{m(\max)}} = \frac{75}{10} = 7.5$$

$$\begin{aligned} \rightarrow \text{System BW} &= 2(f_{m(\max)} + \Delta f) \\ &= 2(10 + 75) \\ &= 170 \text{ kHz} \end{aligned}$$

Example 3. The equation of an angle-modulated voltage is

$$\Rightarrow e = 10 \sin (10^8 t + 3 \sin 10^3 t)$$

What form of angle modulation is this while the message signal is a cos wave? Calculate the carrier and modulating frequencies, the modulation index and deviation and the power dissipated in 100Ω resistor.

→ Standard Form of FM wave

$$\Rightarrow e = V_c \sin (\omega_c t + m_f \sin (\omega_m t))$$

$$- V_c = 10 \text{ Volt}$$

$$\omega_c = 10^8 \text{ rad/sec} \Rightarrow f_c = \frac{10^8}{2\pi} = 15.91 \text{ MHz}$$

$$m_f = 3$$

$$\omega_m = 10^3 \text{ rad/sec} \Rightarrow f_m = \frac{10^3}{2\pi} = 159.1 \text{ Hz}$$

$$\rightarrow m_f = \frac{\Delta f}{f_m} \Rightarrow \Delta f = m_f f_m = 3 \times 159.1 \text{ Hz} = 477.3 \text{ Hz} \quad \left| \quad \rightarrow P = \frac{V_c^2}{2R} = \frac{100}{2 \times 100} = 0.5 \text{ W}$$

Examples of Frequency Modulation

Example 4. Use Carson's Rule to compare the bandwidth that would be required to transmit a baseband signal with a frequency range from 300 Hz to 3 kHz using

- a. NBFM with maximum deviation of 5 kHz.
- b. WBFM with maximum deviation of 75 kHz.

~> For NBFM

$$\begin{aligned}\Rightarrow B_w &= 2(f_m + \Delta f) \\ &= 2(3 + 5) \\ &= 16 \text{ KHz}\end{aligned}$$

~> For WBFM

$$\begin{aligned}\Rightarrow B_w &= 2(f_m + \Delta f) \\ &= 2(3 + 75) \\ &= 156 \text{ KHz}\end{aligned}$$

Example 5. The carrier frequency of an FM broadcast transmitter is 100 MHz, and the maximum frequency deviation is 75 kHz. If the highest audio frequency modulating carrier is 15 kHz. What is the approximate bandwidth of the signal?

$$\rightarrow f_c = 100 \text{ MHz}$$

$$\Delta f = 75 \text{ kHz}$$

$$f_m = 15 \text{ kHz}$$

$$\begin{aligned} \rightarrow BW &= 2(f_m + \Delta f) \\ &= 2(15 + 75) \\ &= 180 \text{ kHz} \end{aligned}$$

Example 6. A single tone FM signal is given by:

$$\Rightarrow V = 10 \sin \left(16\pi \times 10^8 t (+ 3 \sin) 2\pi \times 10^3 t \right) \text{ Volt}$$

Find the modulation index, modulating frequency, deviation, carrier frequency and power of the FM signal.

~> Standard eqⁿ of FM signal

$$\Rightarrow V = V_c \sin (\omega_c t + m_f \sin (\omega_m t))$$

$$\rightarrow V_c = 10 \text{ Volts}$$

$$\omega_c = 16\pi \times 10^8 \text{ rad/sec} \Rightarrow f_c = \frac{\omega_c}{2\pi} = 8 \times 10^8 \text{ Hz}$$

$$\omega_m = 2\pi \times 10^3 \text{ rad/sec} \Rightarrow f_m = \frac{\omega_m}{2\pi} = 10^3 \text{ Hz}$$

$$m_f = 3$$

$$\rightarrow m_f = \frac{\Delta f}{f_m} \Rightarrow \Delta f = m_f f_m = 3 \times 10^3 \text{ Hz}$$

$$\rightarrow P = \frac{V_c^2}{2R} = \frac{100}{2R} = \frac{50}{R} \text{ Watt.}$$

Examples of Frequency Modulation

Example 7. A FM wave is represented by the following equation:

$$\Rightarrow V = 10 \sin \left(5 \times 10^8 t + 4 \sin 1250 t \right) \text{ Volt}$$

Find,

A. Carrier and Modulating Frequencies

B. Modulation Index and Maximum Deviation

C. The Power dissipated by this FM wave in 5Ω resistor.

Standard Form of FM wave:
 $\Rightarrow V = V_c \sin(\omega_c t + m_f \sin(\omega_m t))$

- $V_c = 10 \text{ Volt}$

$$\omega_c = 5 \times 10^8 \text{ rad/sec} \Rightarrow f_c = \frac{\omega_c}{2\pi} = 79.57 \text{ MHz}$$

$$\omega_m = 1250 \text{ rad/sec} \Rightarrow f_m = \frac{\omega_m}{2\pi} = 199 \text{ Hz}$$

$$m_f = 4$$

$$\begin{aligned} \rightarrow m_f &= \frac{\Delta f}{f_m} \Rightarrow \Delta f = m_f f_m \\ &= 4 \times 199 \\ &= 796 \text{ Hz} \end{aligned}$$

$$\rightarrow P = \frac{V_c^2}{2R} = \frac{100}{2 \times 5} = 10 \text{ Watts}$$

Example 8. In a FM System, the modulating frequency $f_m = 1 \text{ kHz}$, the modulating voltage

$E_m = 2 \text{ volt}$ and the deviation is 6 kHz . If the modulating voltage is raised to 4 volt then what is the new deviation? If the modulating voltage is further increased to 8 volt and modulating frequency is reduced to 500 Hz what will be deviation?

$$\Delta f = 6 \text{ kHz}$$

$$\begin{aligned}\Rightarrow K_f &= \Delta f / E_m \\ &= \frac{6}{2} = 3 \frac{\text{kHz}}{\text{V}}\end{aligned}$$

$$\begin{aligned}\rightarrow \text{Case-II, } E_m &= 4 \text{ Volt} \Rightarrow \Delta f = K_f E_m \\ &= 3 \times 4 \\ &= 12 \text{ kHz}\end{aligned}$$

$$\begin{aligned}\rightarrow \text{Case-III, } E_m &= 8 \text{ Volt} \\ f_m &= 500 \text{ Hz} \Rightarrow \Delta f = K_f E_m \\ &= 3 \times 8 \\ &= 24 \text{ kHz}\end{aligned}$$

Example 9. What is the bandwidth required for an FM signal if the modulating frequency is 1kHz and the maximum deviation is 10kHz? What is the BW required for a DSBFC (AM) transmission?

→ FM wave

$$- f_m = 1 \text{ kHz}$$

$$\Delta f = 10 \text{ kHz}$$

$$\begin{aligned} \rightarrow BW &= 2(f_m + \Delta f) \\ &= 2(1 + 10) \\ &= 22 \text{ kHz} \end{aligned}$$

→ DSBFC wave

$$\Rightarrow BW = 2f_m$$

$$= 2 \times 1$$

$$= 2 \text{ kHz}$$

$$\rightarrow (BW)_{FM} \gg (BW)_{AM}$$

Examples of Frequency Modulation

Example 10. A 400 Hz modulating signal modulates a 20 MHz carrier. The carrier voltage is 5V, and the maximum deviation is 10 kHz. Write down the mathematical expression for the FM wave considering the carrier and message signal as sin and cos wave respectively.

$$\Rightarrow f_m = 400 \text{ Hz}$$

$$f_c = 20 \text{ MHz}$$

$$V_c = 5 \text{ V}$$

$$\Delta f = 10 \text{ kHz}$$

$$\Rightarrow m_f = \frac{\Delta f}{f_m} = \frac{10 \times 10^3}{400} = 25$$

$\Rightarrow$ Standard FM wave eqⁿ is given by:

$$y_{FM} = V_c \sin(\omega_c t + m_f \sin \omega_m t)$$
$$= 5 \sin(2\pi \times 20 \times 10^6 t + 25 \sin(2\pi \times 400 t))$$

Example 11. When the modulating frequency in an FM system is 400 Hz and the modulating voltage is 2.4V, the modulation index is 60. Calculate the maximum deviation. What is the modulation index when the modulating frequency is reduced to 250 Hz and the modulating voltage is simultaneously raised to 3.2V?

→ Case-I — $F_m = 400 \text{ Hz}$
 $V_m = 2.4 \text{ V}$
 $m_f = 60$

$$\rightarrow m_f = \frac{\Delta F}{F_m} \Rightarrow \Delta F = m_f F_m$$
$$= 60 \times 400$$
$$= 24 \text{ kHz}$$

$$\rightarrow \Delta F = K_f V_m \Rightarrow K_f = \frac{\Delta F}{V_m} = \frac{24}{2.4} = 10 \frac{\text{kHz}}{\text{V}}$$

→ Case-II — $F_m = 250 \text{ Hz}$
 $V_m = 3.2 \text{ Volt}$

$$\rightarrow m_f = \frac{\Delta F}{F_m} = \frac{K_f V_m}{F_m}$$
$$= \frac{10 \times 10^3 \times 3.2}{250}$$
$$= 128$$

Examples of Frequency Modulation

Example 12. In an FM system, the audio frequency is 1 kHz, and the audio voltage is 2 volts. The deviation is 4 kHz. If the AF voltage is now increased to 8 volts, and its frequency dropped to 500 Hz. Find the modulating index in each case and the corresponding bandwidth using Carson's rule.

Case 1: $F_{m1} = 1 \text{ kHz}$, $V_{m1} = 2 \text{ V}$, $\Delta F_1 = 4 \text{ kHz}$

$$- m_{F1} = \frac{\Delta F_1}{F_{m1}} = \frac{4}{1} = 4$$

$$- BW = 2(F_{m1} + \Delta F_1) \\ = 2(1 + 4) \\ = 10 \text{ kHz}$$

$$\rightarrow \Delta F_1 = K_F V_{m1} \Rightarrow K_F = \frac{\Delta F_1}{V_{m1}} = \frac{4}{2} = 2 \frac{\text{kHz}}{\text{V}}$$

Case 2: $F_{m2} = 0.5 \text{ kHz}$, $V_{m2} = 8 \text{ V}$

$$- \Delta F_2 = K_F V_{m2} \\ = 2 \times 8 = 16 \text{ kHz}$$

$$- m_{F2} = \frac{\Delta F_2}{F_{m2}} = \frac{16}{0.5} = 32$$

$$- BW = 2(F_{m2} + \Delta F_2) = 2(0.5 + 16) = 33 \text{ kHz}$$

Example 13. A message $m(t) = A_m \sin(2\pi f_m t)$ is used to modulate the phase of a carrier $A_c \sin(2\pi f_c t)$ to get the modulated signal $y(t) = A_c \sin(2\pi f_c t + m(t))$.
 a. Depends on A_m but not on f_m
 b. Depends on f_m but not on A_m
 c. Depends on both A_m and f_m
 d. Does not depend on A_m and f_m

c. Depends on both A_m and f_m

$$\Rightarrow \frac{d}{dt} \theta_i(t) = \omega_i(t) = 2\pi f_i(t) = 2\pi f_c + m'(t)$$

d. Does not depend on A_m and f_m

$$[As m(t) = A_m \sin(2\pi f_m t), so m'(t) = A_m(2\pi f_m) \cos(2\pi f_m t)]$$

$$\Rightarrow f_i(t) = f_c + A_m f_m \cos(2\pi f_m t)$$

$$\Rightarrow f_i(t) - f_c = A_m f_m \cos(2\pi f_m t)$$

$f_i(t)$ will be maximum when $\cos(2\pi f_m t) = 1$ and then $(f_i(t)_{max} - f_c)$ is represented as Δf .

$$So, f_i(t)_{max} - f_c = A_m f_m = \Delta f$$

Now, Bandwidth of $y(t)$ is $B = 2(f_m + \Delta f) = 2(f_m + A_m f_m)$ [So, Answer is c]

Example 14. Consider an FM signal:

$$\Rightarrow f(t) = \cos(2\pi f_c t + \beta_1 \sin(2\pi f_1 t) + \beta_2 \sin(2\pi f_2 t)) \text{ Volt}$$

Here $f(t)$ is a multi-tone FM signal with 2 modulation indexes (i.e., β_1 and β_2)

The maximum deviation of instantaneous frequency from the carrier frequency f_c is

$$\sim) \phi = 2\pi f_c t + \beta_1 \sin(2\pi f_1 t) + \beta_2 \sin(2\pi f_2 t)$$

$$\Rightarrow \frac{d\phi}{dt} = 2\pi f_c + \beta_1 (2\pi f_1) \cos(2\pi f_1 t) + \beta_2 (2\pi f_2) \cos(2\pi f_2 t)$$

$$= 2\pi \left(f_c + \underbrace{\beta_1 f_1 \cos(2\pi f_1 t) + \beta_2 f_2 \cos(2\pi f_2 t)} \right)$$

↗
Constant

↓
Freq. deviation.

$$= (\beta_1 f_1 + \beta_2 f_2)_{\max}$$

Example 15. Consider an FM signal:

$$\Rightarrow f(t) = 10 \cos \left(2\pi \times 10^5 t + 5 \sin 2\pi \times 1500 t + 7.5 \sin 2\pi \times 1000 t \right) \text{ Volts}$$

The deviation ratio of given FM signal is

$$\Rightarrow \phi = 2\pi \times 10^5 t + 5 \sin(2\pi \times 1500 t) + 7.5 \sin(2\pi \times 1000 t)$$

$$\Rightarrow \frac{d\phi}{dt} = 2\pi \times 10^5 + 5(2\pi \times 1500) \cos(2\pi \times 1500 t) + 7.5(2\pi \times 1000) \cos(2\pi \times 1000 t)$$

$$= 2\pi \left(\underbrace{10^5}_{f_c} + \underbrace{7500 \cos(2\pi \times 1500 t) + 7500 \cos(2\pi \times 1000 t)}_{\Delta f_{\max}} \right)$$

$$\Delta f_{\max} = 7500 + 7500 = 15000 \text{ Hz}$$

$$\text{Deviation Ratio} = \frac{\Delta f}{f_{m(\max)}} = \frac{15000}{1500} = 10$$

Example 16. An angle modulated signal is given by:

$$\Rightarrow s(t) = \cos \left(2\pi \left(2 \times 10^6 t + 30 \sin(150t) \right) \right) + 40 \sin 150t \text{ Volt}$$

The maximum frequency deviation and phase deviation of above signal are

Step 1: Simplify the signal

Both terms are same frequency:

$$30 \sin(150t) + 40 \sin(150t) = 70 \sin(150t)$$

So,

$$s(t) = \cos \left(2\pi \left(2 \times 10^6 t + 70 \sin(150t) \right) \right)$$

Step 2: Maximum phase deviation

Phase deviation term:

$$\phi(t) = 2\pi \cdot 70 \sin(150t)$$

Maximum value:

$$\phi_{\max} = 2\pi \times 70$$

$$\boxed{\phi_{\max} = 140\pi \text{ rad}}$$

Step 3: Instantaneous frequency

$$f_i(t) = \frac{1}{2\pi} \frac{d\theta(t)}{dt}$$

$$\theta(t) = 2\pi(2 \times 10^6 t + 70 \sin(150t))$$

Differentiate:

$$f_i(t) = 2 \times 10^6 + 70 \times 150 \cos(150t)$$

$$f_i(t) = 2 \times 10^6 + 10500 \cos(150t)$$

Step 4: Maximum frequency deviation

$$\Delta f_{\max} = 10500 \text{ Hz}$$

Example 17. An angle modulated signal is given by:

$$\Rightarrow s(t) = \left(\cos(2\pi \times 10^6 t + 30 \sin(150t)) \right) + 40 \cos(150t) \text{ Volt}$$

The maximum phase deviation is

Step 1: Identify instantaneous phase

Signal are

$$\theta(t) = 2\pi(2 \times 10^6 t + 30 \sin(150t) + 40 \cos(150t))$$

Step 2: Maximum phase deviation

The phase deviation term is:

$$\phi(t) = 2\pi(30 \sin(150t) + 40 \cos(150t))$$

This is of form:

$$A \sin x + B \cos x = \sqrt{A^2 + B^2} \sin(x + \phi)$$

So maximum value:

$$\phi_{\max} = 2\pi\sqrt{30^2 + 40^2} = 2\pi\sqrt{900 + 1600} = 2\pi\sqrt{2500} = 2\pi \times 50$$

$$\boxed{\phi_{\max} = 100\pi \text{ rad}}$$

Step 3: Instantaneous frequency

$$f_i(t) = \frac{1}{2\pi} \frac{d\theta(t)}{dt}$$

Differentiate:

$$\frac{d\theta}{dt} = 2\pi \left(2 \times 10^6 + 30 \cdot 150 \cos(150t) - 40 \cdot 150 \sin(150t) \right)$$

$$f_i(t) = 2 \times 10^6 + 4500 \cos(150t) - 6000 \sin(150t)$$

Step 4: Maximum frequency deviation

Again combine:

$$4500 \cos(150t) - 6000 \sin(150t)$$

Amplitude:

$$\Delta f_{\max} = \sqrt{4500^2 + 6000^2} = \sqrt{20.25 \times 10^6 + 36 \times 10^6} = \sqrt{56.25 \times 10^6} = 7500 \text{ Hz}$$